TEMASEK JUNIOR COLLEGE
Preliminary Examinations 2009

BIOLOGY	**9747/02**
Higher 2	**Thursday, 17 September**
Paper 2 Core Paper	**2 hours**

Question 1 [7 marks]

The electronmicrograph in Figure 1.1 shows an acinar cell found in the pancreas.

Figure 1.1

(a) (i) With reference to the Fig. 1.1, suggest the principle function of the pancreatic acinar cell. Give **two** reasons to explain your answer. [3]

- **secretion of pancreatic / digestive enzymes**

or

- **secretion of hormones [1]**

(Any two)

- **presence of large amount of rER, which is the site where pancreatic enzymes are synthesized**

  or

  presence of sER, which is the site where steroid hormones / lipids are synthesized
- **presence of large number of secretory vesicles for release of pancreatic enzymes outside the cell [1]**
- **presence of mitochondria which synthesizes ATP for**
  - **Activation of amino acids (giving amino acid-AMP)** OR
  - **Formation of peptide bonds (at ribosomes)** OR
  - **Transport / Movement of vesicles towards cell surface membrane for release of pancreatic enzymes [1]**

Figure 1.2 shows two different structures, A and B, found in an animal cell.

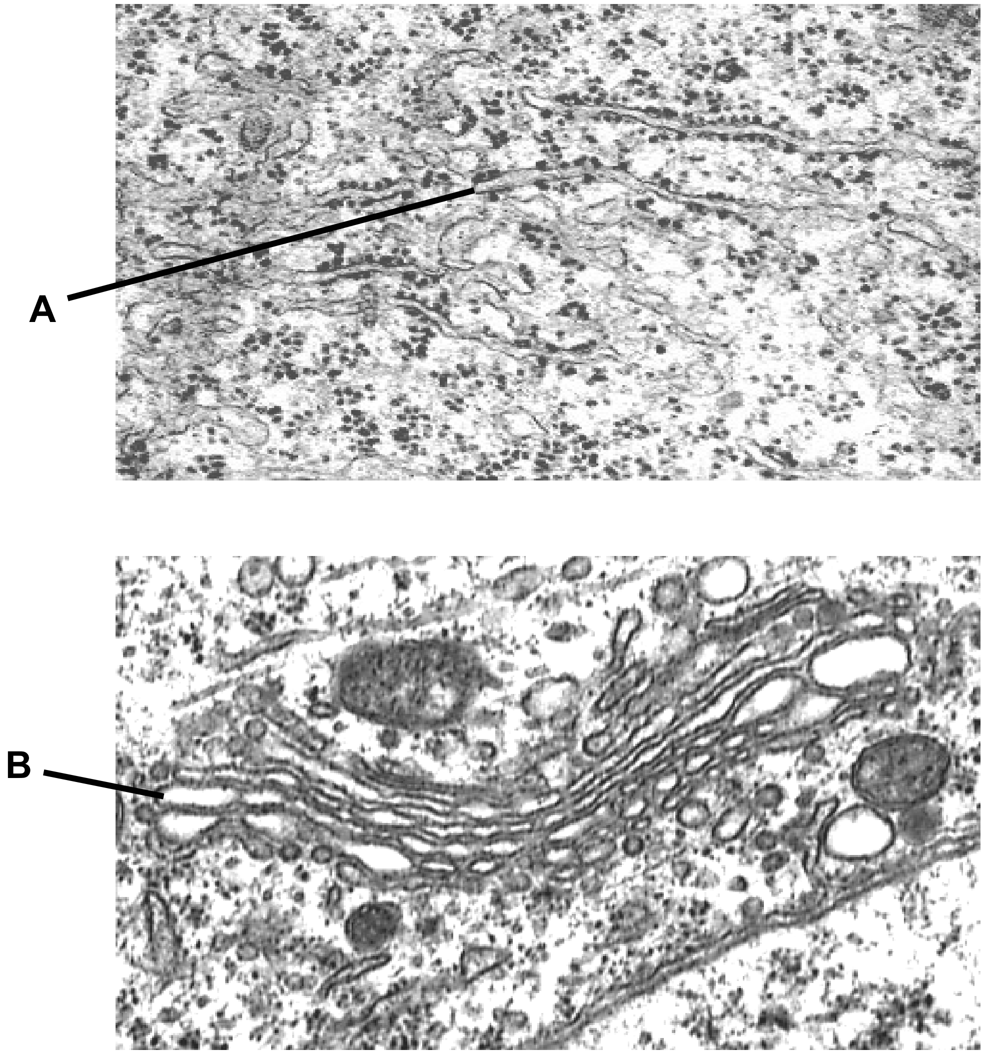


Figure 1.2

(b) (i) Describe **three** ways in which A is similar to B. [3]

- **Both are bound by a single membrane / single membrane-bound structures. [1]**
- **Both consist of membranous sacs called cisternae, in which newly synthesized proteins undergo processing. [1]**
- **Both give rise to membrane-bound vesicles which are involved in transport of newly synthesized proteins within the cell. [1]**
- **Both are involved in protein synthesis / secretion. [1]**

(ii) The products of B are usually in the inactive form. Suggest the significance of this. [1]

- **to prevent the enzymes from digesting proteins / contents within the cells in which they are synthesized. [1]**

Question 2 [Total: 12 marks]

Photomicrographs in Figure 2.2 show the stages of mitosis in a bluebell (Figure 2.1), which has a diploid number of 16.

Cells undergoing interphase [1]

Figure 2.1

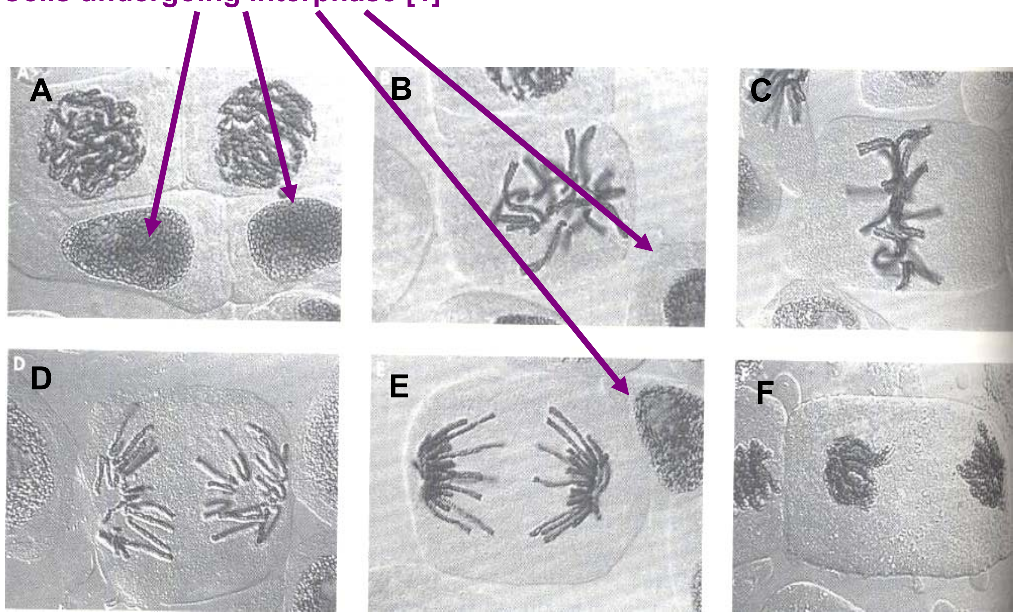


Figure 2.2

(a) Identify using arrow(s) and label on Figure 2.2, which cells are undergoing interphase and explain your answer. [2]

[any two cells of shown above]

- **Chromosomes [not visible as separate structures/ in uncondensed state]**
- **Presence of nuclear envelope**

(b) With reference to Figure **2.2 C**,

(i) state the stage of division observed. [1]

- **Metaphase**

(ii) draw and label cells undergoing a similar stage during formation of gametes, in the space below. [2]

Marking points:

- **Metaphase I – 8 bivalents/ pairs of homologous chromosomes aligned at equator**
- **Metaphase II – 2 daughter cells; 8 chromosomes in each cell; aligned at equator; equator perpendicular to that drawn for Metaphase I**

Note: penalize for drawing/labeling centrioles since this is a plant cell (max 1 mark with everything else correct)

(c) Explain the significance of synapsis during gamete formation. [3]

- **Pairing of <u>homologous chromosomes</u> to form <u>bivalents</u>**
- **During <u>Prophase I</u>, for crossing over between <u>non-sister chromatids</u>**
- **<u>New combinations of alleles</u> on chromosomes; creates <u>genetic variation</u>**

Figure 2.3 shows changes in DNA content at different stages of the animal life cycle.

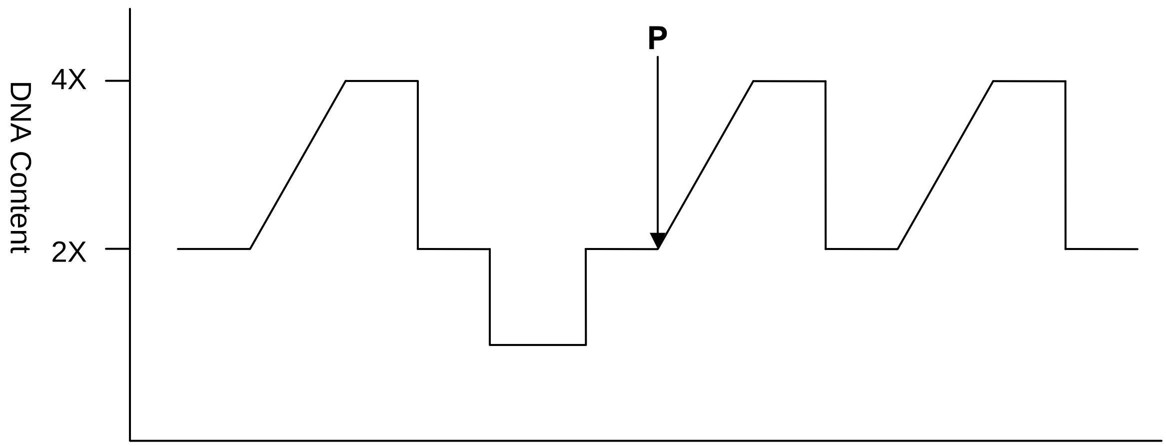


Figure 2.3

(d) With reference to Figure 2.3,

(i) identify and describe Stage P. [3]

- **<u>DNA replication</u> during <u>Interphase</u>**
- **Doubling of DNA content from <u>2X to 4X</u>**
- **2 parental DNA strands act as <u>templates</u> for synthesis of <u>complementary</u> daughter strands**
- **<u>DNA polymerase</u>; new strand formed in <u>5' to 3' direction</u>; <u>proofreading</u>**

(ii) state the significance of Stage P. [1]

- **to produce genetically identical daughter cells**

Question 3 [13 marks]

In prokaryotes, genes are often organized into operons. Some examples of well-studied operons include the *lac* and *trp* operons, which are involved in lactose metabolism and tryptophan synthesis respectively.

(a) Explain the meaning of the term 'operon'. [3]

- **Unit of prokaryotic gene expression and regulation, consisting of multiple, [contiguous/ OWTTE] structural genes and control elements. [1]**
- **Consist of genes that encode enzymes of [a single biochemical pathway/ with related functions]. [1]**
- **Expression of genes on an operon controlled by same promoter and operator. [1]**

(b) Discuss the similarity between the control of the *trp* and *lac* operons. [2]

- **Both are under negative control.**
  - **When the active repressor binds the operator. [1]**
- **Binding of RNA polymerase to the promoter is blocked, resulting in genes on the operon not being [expressed/ transcribed]. [1]**

(c) With the use of named examples, discuss the differences between inducible and repressible operons. [4]

	Inducible operon	Repressible operon
Named example	Lac operon	Trp operon
Differences in regulation	Usually [not expressed/ turned off], but can be [stimulated/ induced] when an inducer molecule [binds/ interacts] with a regulatory protein.	Usually [turned on/ expressed], but can be [inhibited/ repressed] when a [specific small molecule/ OWTTE] binds allosterically to its regulatory protein
Gene product of regulatory gene	Synthesizes active repressor	Synthesizes inactive repressor
Events when lactose/ tryptophan is absent	Active repressor binds operator, [transcription turned off/ operon not expressed]	Inactive repressor cannot bind operator, [transcription turned on/ operon expressed]
Events when lactose/ tryptophan is present	[Allolactose/ inducer] binds repressor, repressor becomes inactive	[Co-repressor/ tryptophan] binds repressor, repressor becomes active
	Inactive repressor cannot bind operator, [transcription turned on/ operon expressed]	Active repressor binds operator, [transcription turned off/ operon not expressed]

A certain temperate bacteriophage is known to insert its own genome within the *trp* operon, downstream of the promoter. The genome of the bacteriophage contains excision genes. When expressed, these genes will result in the excision of the prophage from the host genome.

(d) Explain how changes in the amount of tryptophan present in the immediate vicinity of bacteria infected by the phage might affect the numbers of free bacteriophage detected. [4]

Description of what will happen

- **No. of free bacteriophages <u>increase when trp is absent</u> and <u>[stay the same/ decrease] when trp is present.</u> [1]**

Explanation (max 3m)

Trp absent

- **<u>[Operon/ prophage within operon]</u> is <u>[expressed/ turned on/ activated]</u> due to absence of trp [1]**
- **<u>Enzymes</u> that excise prophage are <u>synthesized, prophage is excised</u> and enters <u>lytic cycle.</u> [1]**

Trp present

- **<u>[Operon/ prophage within operon]</u> is <u>[not expressed/ turned off/ inactivated]</u> due to presence of trp. [1]**
- **Thus <u>enzymes</u> that excise prophage <u>not synthesized</u>, <u>prophage not excised</u> and stays in <u>lysogenic cycle.</u> [1]**

<u>Question 4</u> [11 marks]
Figure 4.1 shows an electron micrograph of *Escherichia coli*.

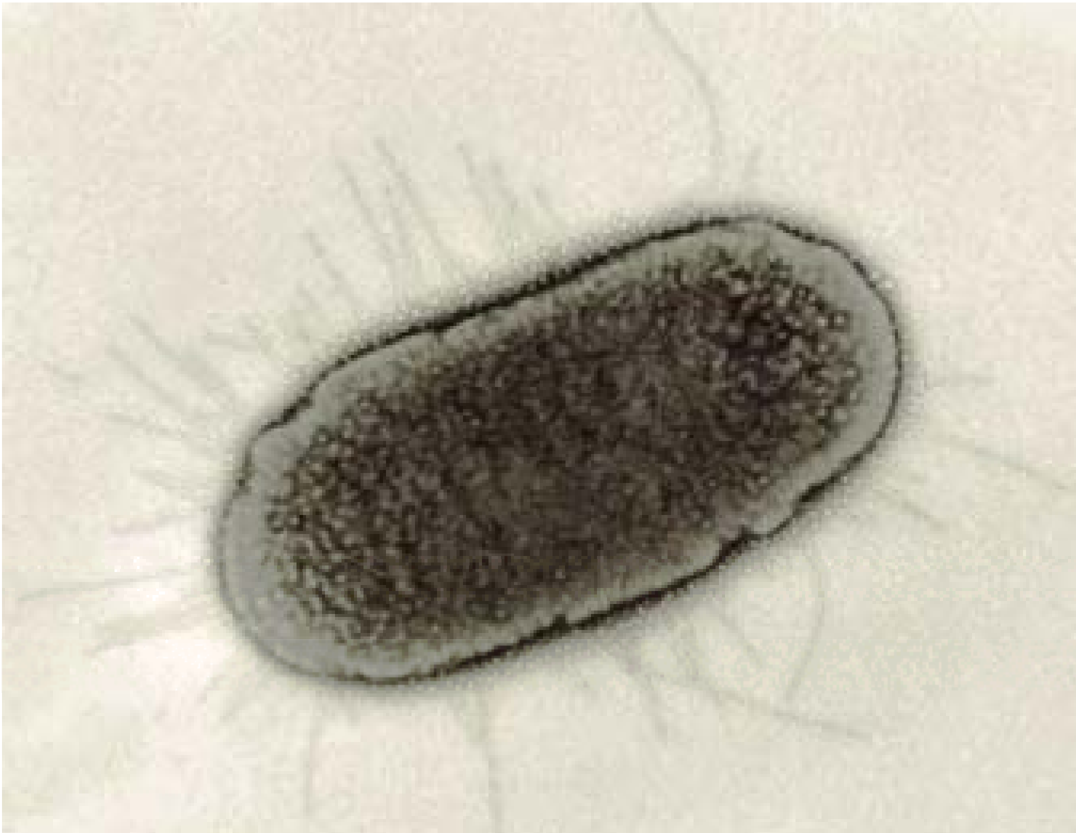


Figure 4.1

(a) Describe the structural organization of DNA in a bacterial cell. [3]

- **Bacterial chromosome consists of one main <u>circular covalently closed double stranded DNA molecule</u>. [1]**
- **Presence of one or more <u>extrachromosomal plasmids</u>, which <u>replicate independently of the bacterial chromosome</u>. [1]**
- **The DNA molecule exists in a <u>highly condensed</u> configuration, and is organized into <u>supercoiled looped domains</u>. [1]**
- **Numerous proteins called <u>nucleoid-associated proteins</u>, fold the bacterial chromosome into a <u>compact structure</u>. [1]**
- **Example: <u>H-NS protein</u>, a major DNA-binding protein which binds to DNA and compacts DNA structure.**

Bacteria reproduce by a process called binary fission.

(b) (i) Explain the significance of binary fission in bacteria replication. [4]

- **Binary fission results in the production of <u>genetically identical daughter cells</u>. [1]**

- **Replication of bacteria chromosome occurs during binary fission. During DNA replication, the double-stranded DNA molecule is separated and each strand acts as a template for the synthesis of daughter strands. [1]**
- **DNA synthesis progresses bidirectionally around the circular chromosome, resulting in the production of two identical DNA molecules. [1]**
- **As the cell grows and elongates, the newly replicated DNA molecules are actively separated within the cell. A septum then grows and splits the cell into two genetically identical daughter cells. [1]**

- **Binary fission is a form of asexual reproduction in bacteria, and is a fast process. This allows rapid reproduction of bacteria. [1]**

- **Binary fission results in the production of genetically identical daughter cells, thus genes that code for desirable traits are passed on to daughter cells, thus favouring successful colonization / survival of a species. [1]**

(ii) Distinguish between binary fission and mitosis. [2]

(Any two)

- **Binary fission is the process by which prokaryotic cells reproduce asexually, whereas mitosis is the process by which the nucleus of a eukaryotic cell divides to produce genetically identical daughter cells. [1]**
- **Mitosis occurs only in eukaryotic cells, whereas binary fission occurs can occur in both prokaryotic and eukaryotic cells. [1]**
- **In mitosis, chromosomes are attached to spindle fibres at their centromeres. However in binary fission, the chromosome is attached to the bacterial cell membrane during DNA replication. [1]**
- **Mitosis involves structures such as spindle fibres, centrioles, centromeres and kinetochores. However, binary fission does not require these structures. [1]**

Bacteria can reproduce at extremely rapid rates. An *E.coli* cell is capable of dividing every 20 minutes when grown under optimal conditions. However, experiments have shown that the time taken to completely replicate one chromosome is often longer than the average time taken for the cell to divide.

(c) (i) Suggest how it is possible for a bacterium to divide at a rate faster than that at which its DNA can be replicated. [1]

- **During DNA replication, multiple replication bubbles can initiate at the origin of replication. [1]**

(ii) Given that the E.coli chromosome contains 4 million nucleotides, and that the rate of DNA replication occurs at 1000 nucleotides per second, calculate the time required for an entire *E.coli* chromosome to be replicated. [1]

- **4 000 000 / (1 000 x 2) = 2 000 s [1]**

Note: Students should remember that DNA replication occurs bidirectionally in bacteria.

Question 5 [12 marks]

Three different feather colours have been observed in Andalusian fowl – black, white and blue. Comparison of the alleles for white feathers and black feathers showed that there was a deletion of a single base-pair in the allele coding for white feathers.

To determine the mode of inheritance of feather colour in these fowl, several types of crosses were carried out.

In the parental generation, 2 crosses were carried out:
1. Female, pure-breeding black feathered fowl x male, pure-breeding white feathered fowl
2. Male, pure-breeding black feathered fowl x female, pure-breeding white feathered fowl

All of the F_1 generation in both crosses had blue feathers.

Sibling mating was then carried out within the F_1 generation, and the following results were obtained:

Black feathered fowl: 439
White feathered fowl: 442
Blue feathered fowl: 879

(a) State the type of cross carried out in the parental generation. [1]

- **Reciprocal cross [1]**

(b) Explain why a reciprocal cross was **not** carried out within the F_1 generation. [2]

- **Trait is not sex-linked/ is autosomal/ OWTTE [1]**
- **As phenotypic ratio of F_1 generation was the same for both parental crosses. [1]**

(c) Using suitable symbols, construct a genetic diagram to explain the results of the F_1 sibling mating. [5]

Mark scheme:
Correct key: 1m
Correct F_1 genotype and phenotype: 1m
Correct gametes: 1m
Correct F_2 phenotype, genotype and ratio, 2m

Key: B represents the allele for black feathers
W represents the allele for white feathers
B is incompletely dominant over W (Reject co-dominance)

F1 phenotype:	**Blue feathers**		**x**		**Blue feathers**
F1 genotype:	**BW**		**x**		**BW**
Gametes:	**½ B**	**½ W**		**½ B**	**½ W**
F2 genotype:	**BB**		**2 BW**		**WW**
F2 phenotype:	**Black feathers**	**:**	**Blue feathers**	**:**	**White feathers**
Ratio:	**1**	**:**	**2**	**:**	**1**

For H2

Fireflies contain a compound called luciferin, which can be oxidised by the enzyme luciferase to oxyluciferin. Light is emitted in the process of this reaction.

Researchers have discovered that D-cystine and 2-cyano-6-hydroxybenzothiazole undergo a condensation reaction to form luciferin. This reaction is mediated by enzyme A.

To study the genetics of light emission in fireflies, some researchers have bred mutant fireflies that have mutations that result in non-functional enzyme A and luciferase. These mutant fireflies do not emit light.

The alleles A and L code for functional enzyme A and luciferase respectively, while the alleles a and l code for non-functional enzyme A and non-functional luciferase respectively.

Figure 5.1 shows biochemical pathway that eventually results in the emission of light in fireflies.

$$\boxed{\text{D- cystine + 2-cyano-6-hydroxybenzothiazole}} \xrightarrow{\text{Enzyme A}} \boxed{\text{Luciferin}} \xrightarrow{\text{Luciferase}} \boxed{\text{Oxyluciferin + light}}$$

Fig 5.1

Pure-breeding normal fireflies were crossed with pure-breeding mutant fireflies. The F_1 generation were all phenotypically normal and could emit light. The F_1 generation was then sibling mated, and the results of the sibling mating are as follows:

Light emitting fireflies: 169
Mutant fireflies: 131

(d) (i) State the term used to refer to this type of gene interaction. [1]

- **Complementary gene action [1]**

(ii) Explain how this type of gene interaction can result in the experimental ratios observed for the sibling mating of the F_1 generation. [4]

- **Functional copies of both enzyme A and luciferase are required for light emission in fireflies. [1]**
- **Alleles A and L are dominant, thus the genotypes A-L- will result in functional copies of both enzyme A and luciferase being formed, thus fireflies with the above genotype will emit light/ OWTTE [1]**
- **The genotypes aa-- and --ll will result in non-functional enzyme A or non-functional luciferase being synthesized, thus the firefly will not emit light/ OWTTE [1]**
- **In a cross between 2 heterozygotes, 9/16 of the offspring will have the genotype A-L-, and 7/16 of the offspring will have the genotypes aa-- or –ll, resulting in the experimental results observed above. [1]**

Question 6 **[Total: 12 marks]**

The Orchid family is the largest family of flowering plants. Since the 19th century, horticulturists have produced more than 100 000 hybrids and cultivated varieties. Orchids are cosmopolitan, occurring in almost every habitat apart from deserts and glaciers. Some orchids are saprophytes and they have no need for leaves, as all their nutrition is gained from dead vegetation and animal material. They only appear above ground to flower and seed. One such species is the Ghost Orchid *Epipogium gmelini*, found only in the deepest beech woods. Another is the Bird's Nest Orchid, which at first glance, can be easily mistaken for a member of the Broomrape family Orobanchaceae, where all are parasitic and feed from the roots of other plants.

Orchid conservationists have so hard a time trying to germinate orchid seeds and grow orchids in an artificial environment. Orchid seeds are tiny, frail and delicate that contain virtually nothing except an embryo. They must enter symbiotic relationship with various fungi that provide them the necessary nutrients to germinate. Orchids have developed highly specialized pollination systems. In some extremely specialized orchids, the flower is adapted to have a colour, shape and odour which attracts male insects via mimicry of a receptive female. Pollination happens as the insect attempts to mate with flowers.

(a) Explain if the biological species concept is suitable to study the Orchid family. [4]

- Members of a biological species are **reproductively compatible**/ able to interbreed; to **produce viable, fertile offspring**
- Orchids are able to produce **many forms of hybrids** and cultivated varieties
- **Limited reproductive isolation** between orchid species
- Thus biological species concept is **not suitable** for study of the Orchid family

(b) Between the morphological and the ecological species concepts, suggest which will be more useful for research on the Bird's Nest orchid. [3]

- **Ecological** species concept

- Morphological species concept groups individuals based on **physical resemblances and differences**
- Bird's Nest orchid **looks similar to a member of the Broomrape family** (thus morphological species concept not as useful)

- Ecological species are **adapted to a particular set of resources**/ ecological niche
- Bird's Nest orchid can be distinguished from Broomrape family as it is **saprophytic**, whereas the latter is **parasitic**

(c) State **one** advantage and **one** disadvantage of using an artificial classification as compared to a natural classification in taxonomy. [2]

- Advantage: use **very simple and easily observable** features, thus **easier to construct**
- Disadvantage: use **only external features**, therefore **not as reliable/** OWTTE

(d) With reference to the passage, state **two** advantages of being saprophytic. [2]

(any two:)

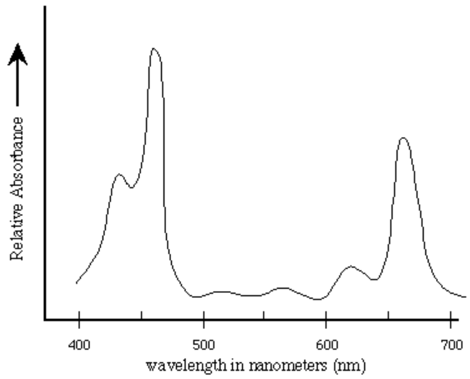
- **No need for leaves** – conserve/ lower need for **resources**
- Found in **deepest beech woods or underground**, where most other plants are unable to survive – **reduced competition**
- **Reduced transpiration**/ water loss since they do not have leaves
- **Do not need sunlight**/ light to survive

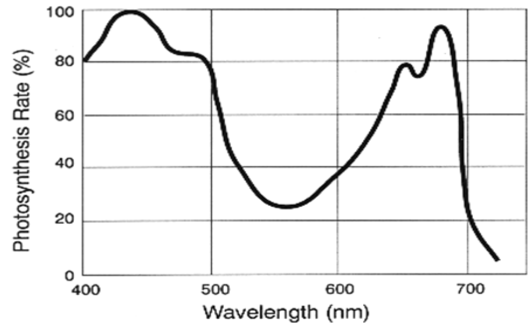
(e) Suggest why orchid conservationists have difficulty germinating orchid seeds. [1]

- **Absence of fungi** in the culture medium to provide **necessary nutrients**/ OWTTE

Question 7 [12 marks]

Fig. 7.1 shows the absorption spectrum of chorophyll a and Fig. 7.2 shows the action spectrum for photosynthesis.





(a) With reference to figures 7.1 and 7.2, explain the role of light harvesting complex in both photosystems I and II. [3]

- **photosynthesis occurs between <u>400 nm to 700 nm</u> / photosynthetically active radiation consist of 400 nm to 700 nm. [1]**
- **The reaction centre of photosystems I and II consists of primary pigments / special chlorophyll a pigments which absorb wavelengths of up to 700 nm and 680 nm respectively unable to absorb light throughout the spectrum of light where photosynthesis occurs. [1]**
- **The antenna / accessory pigments of the <u>LHC absorbs light throughout the PAR and channel the energy towards the RC</u>.[1]**

(b) Distinguish the electron transport chain in photophosphorylation from that in oxidative phosphorylation. [3]

Photophosphorylation	Oxidative Phosphorylation
• accepts electrons from PS I or PS II / special chlorophyll a_{680} or a_{700}	• accepts electrons from NADH and FADH
• ultimately transfers the electron to either PSI or $NADP^+$	• ultimately transfers the electron to O_2 to from water
• energy released from ETC is used to pump H^+ from stroma to thylakoid space	• energy released from ETC is used to pump H^+ from matrix to intermembrane space.

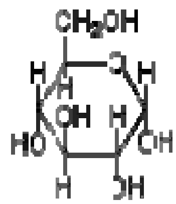
The ATP and NADPH synthesized during photophosphorylation is utilised in the Calvin cycle for the synthesis of glucose, which is subsequently oxdised by the cell to produce ATP for it biochemical processes. Glucose may also converted to sucrose and transported to the the root to be stored as starch.

(c) Suggest a reason why ATP generated in the chloroplast is **not** used directly for the cell's biochemical processes. [1]

- **Chloroplasts do not have channels that allow them to export ATP to the cytosol / ATP obtain from celllular respiration.**

(d) (i) In the space given, draw a monomer of starch. [1]

Ans:



Accept simplified answer.

(ii) Distinguish this molecule from the monomer of cellulose. [1]

- **In α-glucose, which found in starch, the <u>–OH group of carbon 1 project below the plane of the ring</u>, whereas in β-glucose, which is found in cellulose, the <u>–OH group of carbon 1 projects above the plane of the ring</u>**

In animals, triglycerides serves as a long term storage of energy, rather than carbohydrates.

(e) Suggest why triglycerides are better storage molecules than carbohydrates in animals. [3]

- **<u>For an equivalent amount of energy stored, triglycerides possess half the mass of carbohydrates</u>. [1]**
- **Unlike plants, animals move, <u>locomotion requires mass to be kept to a minimum.</u> [1]**
- **When compared to carbohydrates of the same mass, <u>the ratio of energy storing C- H bonds in triglycerides is more than twice that of carbohydrates</u>, making <u>triglycerides much more efficient molecules for storing chemical energy</u> [1]**

Essay Question 8 (a)

Describe how eukaryotic cells can regulate gene expression at the translational and post-translational levels. [4]

Translational level ***(max 2m)***

- **Binding of regulatory proteins to the 5' UTR/ leader sequence/ 5' cap of mRNA, preventing ribosome attachment [1]**
- **Availability of regulatory proteins that are required for translation to take place efficiently [1]**
- **Availability of amino acids can affect the rate of elongation. [1]**
- **Binding of regulatory proteins to poly(A) tail, eventually preventing binding of ribosomes to 5' end [1]**
- **Presence of uORFs (upstream open reading frames) that can decrease efficiency of translation by ribosomes. [1]**

Post-translational level ***(max 2m)***

- **Chemical modification by:** ***(max 1m for chemical modification)***
    - **Phosphorylation (activates/ deactivates protein depending on type of protein) [1]**
    - **Glycosylation - addition of one or more sugar monomers to form glycoproteins. [1]**
    - **Ubiquitination – addition of a ubiquitin (a small protein) which identifies the protein for degradation. [1]**
- **The faster a protein is degraded, the lower the level of expression of the gene coding for that particular protein. [1]**
- **Modification of sequence of amino acids in the polypeptide chain – inhibitory portions of the polypeptide chain can be removed by proteases.**

Essay Question 8 (b)

Describe how an action potential is propagated along an unmyelinated axon. [6]

- **From the dendron / dendrite, the action potential spreads along the cell body towards the axon terminal as a wave of depolarizations followed by repolarizations.**
- **At the point of stimulation, all voltage-gated Na^+ channels open. Na^+ ions rush into the neuron and diffuse laterally from the point of entry. [1]**
- **The immediate neighboring region of the membrane further down the axon is depolarized to threshold potential, causing the next action potential to be initiated there. [1]**
- **In the previous region, voltage-gated K^+ channels open while voltage-gated Na^+ channels close. Efflux of K^+ causes the inside of the membrane to become more negative relative to the outside, resulting in repolarization of the membrane. [1]**
- **Outward movement of K^+ continues as K^+ channels remain open, resulting in hyperpolarization of the membrane. [1]**
- **This constitutes the refractory period – the membrane is insensitive to depolarization. Thus the action potential is propagated in one direction / OWTTE. [1]**
- **This repeated depolarization of adjacent regions results in action potential being propagated along the axon till the terminal is reached. [1]**

Essay Question 8 (c)

Discuss, with examples, how gene or chromosomal mutations in may eventually lead to cancer. [10]

Cancer as a multi-step process

- **Cancer is a multi-step process/ occurs due to an accumulation of mutations [1]**
- **Mutations resulting in the appearance of at least [1 active oncogene/ OWTTE], the loss of function of several tumour suppressor genes [1]**
- **and activation of the gene for telomerase need to take place for cancer to occur. [1]**

Normal functions of proto-oncogenes and tumour suppressor genes

- **Proto-oncogenes code for proteins that regulate/ stimulate cell division, tumour suppressor genes code for proteins involved in [inhibition of cell division/ preventing inappropriate cell cycle progression/ OWTTE] [1]**

Gain of function (GOF) and loss of function (LOF) mutations in each type of gene

- **In proto-oncogense, gain of function mutations OR mutations that result in [increased activity/ increased expression] of the gene, will cause increased stimulation of cell division [1]**
    - ***Which can eventually lead to uncontrolled cell division***
- **Mutations that cause a loss of function in both copies of a tumour suppressor gene will result in the cell losing its ability to prevent uncontrolled cell division/ OWTTE. [1]**
- **GOF mutations occur in a dominant manner, LOF mutations occur in a recessive manner.**

Gene/ chromosomal mutations that could result in a GOF/ LOF mutation

- **Substitution, inversion, deletion, insertion [1] *(at least 3 of 4 must be mentioned in ans)* of nucleotides in a gene/ control element.**
    - **Reject above answer if student mentions frameshift mutation of nucleotide is inserted to/ deleted from non-coding segments (e.g. control elements)**
- **Chromosomal deletion/ inversion/ translocation/ duplication of a segment containing the proto-oncogene/ tumour suppressor gene. [1]**

(credit the above 2 points only once in answer)

Some gene/ chromosomal mutations that result in a gain of function mutation in proto-oncogenes include (max 2m):

Gene mutations

- Point mutations could occur in protein-coding region of proto-oncogene that might result in protein that is [hyperactive/ resistant to degradation/ binds its substrate with increased affinity] [1]

- Mutations in promoter/ control elements of proto-oncogene could also lead to increased expression of the gene. [1]

Chromosomal mutations

- Chromosomal translocation of the proto-oncogene might result in proto-oncogene being moved to a region with [an active promoter/ closer to an enhancer/ any other valid point], resulting in increased expression of the proto-oncogene. [1]

- [Amplification/ duplication] of the entire gene might result in extra copies of the proto-oncogene being formed, resulting in [increased synthesis of the gene product/ increased expression of the gene]. [1]

and constantly stimulate cell division.

✗ Reject demethylation of DNA as these are not gene/ chromosomal mutations.

Some gene/ chromosomal mutations that result in a loss of function mutation in tumour suppressor genes include (max 2m):

Gene mutations

- Point mutations could occur in protein-coding region of tumour supressor gene that might result in protein that is [inactive/ non-functional/ more prone to degradation/ binds to its substrate with less affinity/ any other valid point]. [1]

- Mutations in promoter/ control elements of tumour suppressor gene could also lead to decreased expression of the gene. [1]

Chromosomal mutations

- Chromosomal translocation of the tumour suppressor gene might result in tumour suppressor gene being moved to a region with [no promoter/ less active promoter/ closer to a silencer/ very tight packing/ in heterochromatin/ any other valid point], resulting in decreased expression of the tumour suppressor gene. [1]

- Deletion of the entire/ part of the chromosomal segment containing the tumour suppressor gene, resulting in no expressiion of the tumour suppressor gene. [1]

Leads to tumour suppressor gene being unable to inhibit inappropriate cell division/ cell cycle progression.

✗ Reject methylation of DNA as this is not a gene/ chromosomal mutation.

Essay Question 9 (a)

Explain how the neutral theory of molecular evolution appears to be an argument against Darwin's theory of natural selection. [6]

- **Neutral theory of molecular evolution suggests that most of the genetic variation in populations is the result of mutation and genetic drift but not natural selection. And that genetic drift plays a major role in evolution.** 1

- **The theory focuses on the molecular level such as nucleotide substitutions in DNA and amino acid substitutions in proteins.** 1

- **According to the neutral theory, when genomes of each type of species are being compared, most of the molecular differences are selectively "neutral."**

- **This means that different alleles of a gene do not influence the fitness of the individual organism.**

- **Therefore variations which arose due to different alleles are not subjected to natural selection. This means that variation is neutral.** 2

- **However, according to Darwin's theory, among a variety of individuals, there are some which are better able to withstand the prevailing conditions. These individuals are said to be at selective advantage.**

- **According to the neutral theory, most evolutionary change is the result of genetic drift acting on neutral alleles. It is non-adaptive.**

- **A new allele arises typically through the spontaneous mutation of a single nucleotide within the sequence of a gene.**

- **Through genetic drift these new alleles may be lost or become ["fixed" /permanent].**

- **When an allele carrying a new substitution becomes fixed, a new allele is added to the population. In this way, neutral substitutions tend to accumulate, and genomes tend to evolve.** 2

- **On the other hand, variation within a population is an essential pre-condition for evolution through natural selection.**

- **Variations in this case arise spontaneously, but Darwin made no reference to mutation.**

- **Those better adapted individuals are selected by the environment or nature, and survive to reproductive age to produce offspring similar to themselves.**

Essay Question 9 (b)

Using a named example, explain why a population is the smallest unit that can evolve. [5]

- **Natural selection occurs through interactions between individual organisms and their environment, but individuals do not evolve.** 1

- **Natural selection acts on individuals and their characteristics affect their chances of survival and reproductive success and together with gene flow and genetic drift changes allele frequencies in populations.** 1

- **Evolution can only be measured as changes in relative proportions of heritable variations in a population over a succession of generations.** 1

- **E.g. [marine snails/ any other valid example]**
  - **They have different coloration which represents genetic variations within that population.**
  - **If the predators prefer snails with a particular coloration, that proportion of individuals will probably decline from one generation to the next – these snails produced fewer offspring.** 2
  - **The proportions of snails with other coloration will increase within the overall population.**
  - **Therefore it is the population of marine snails, not individuals which evolve.**

Essay Question 9 (c)

Describe the homeostatic regulation of blood glucagon level. [8]

- **Negative feedback is a principle of homeostasis. [1]**
- **When there is any deviation from a set reference point, mechanisms are set into play, which serve to eliminate that deviation.[1]**
- **In the case of controlling glucagon levels, glucagon secretion is triggered by an decrease in blood glucose levels [1]**
- **Glucagon is then secreted by the α-cells of the islets of Langerhans in the pancreas, resulting in and increase in glucagon levels above normal levels [1]**
- **This brings about an increase in the blood glucose levels back to normal. [1]**
- **This is then detected by α-cells of the islets of Langerhands and via a negative feedback mechanism, the secretion of glucagon by the α-cells is stopped / switched off. [1]**
- **The circulating glucagon is destroyed in the liver or removed in the kidney. [1]**
- **Hence, plasma glucagon levels are returned to normal.[1]**

END OF ANSWERS